INCH-POUND
MIL-STD-3088(AV)
24 SEPTEMBER 2026

DEPARTMENT OF DEFENSE
DESIGN CRITERIA STANDARD

REQUIREMENTS FOR THE DEVELOPMENT OF EQUIPMENT TO SUPPORT LIVE EXTERNAL ATTACHED LOADS

AMSC N/A FSC 4240

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FOREWORD

1. This standard is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

2. This military standard establishes the requirements for the design and testing requirements of Live External Attached Load (LEAL) devices for the transportation of human cargo by Department of Defense (DoD) helicopters. Per Army Regulation AR 70-47 and Department of Defense Instruction DoDI 4540.7, the engineering agency assigned responsibility for certifying all DoD helicopter external loads is the U.S. Army Combat Capabilities Development Command Soldier Center (DEVCOM SC). Per Army Regulation AR 70-62, U.S. Army Aviation and Missile Command (DEVCOM AvMC) has the authority for granting airworthiness releases. Both agencies must concur that the designed equipment meets the requirements specified herein for this equipment to be approved for use.

3. This standard is applicable to all military materiel designed and developed by all DoD agencies accepting it for use.

4. LEAL is a mode of transportation by which human cargo is attached via the exterior of DoD aircraft for the purpose of transporting. This mission is intended to be short range, delivering the occupants to safety via the shortest flight practical. LEAL offers many advantages which include:

a. Time critical insertion/extraction of multiple individuals.
b. Insertion/Extraction from terrain inaccessible to aircraft landing, such as confined spaces and over water.
c. The ability to reach areas inaccessible by ground transportation.
d. Faster transit times than conventional ground transportation.
e. Independence from ground transportation conditions (i.e., congestion, flooding, terrain, fire).
f. LEAL can be expanded beyond the transport of live loads to traditional payloads.

5. The requirements of this military standard are intended to be consistent with the requirements of MIL-STD-209, Interface Standard for Lifting and Tie-down Provisions, MIL-STD-913, Requirements For The Certification Of Sling Loaded Military Equipment For External Transportation By Department Of Defense Helicopters, MIL-L-37957, Military Specification, Litters, Rigid, Stokes, MIL-R-8592/3, Military Specification, Survivors, Sling, Rescue Helicopter Equipment.

6. Comments, suggestions, or questions on this document should be addressed to: U. S. Army Combat Capabilities Development Command, Aviation and Missile Center, Systems Readiness Directorate, Utility Division (DEVCOM AvMC SRD Utility), Redstone Arsenal, AL. 35808. Since contact information can change, you may want to verify the currency of this address information using Acquisition Streamlining and Standardization Information System (ASSIST) at https://assist.dla.mil/.

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1. SCOPE

1.1. Scope. This standard covers the design, testing and performance requirements of equipment for Live External Attached Loads (LEAL) by Army, Navy, Marine Corps, and Coast Guard aircraft. The complete transportability requirements for equipment not specified herein are specified in the individual equipment specification. Appendix A contains the offices of each DoD component that has adopted this standard. The complete air transportability requirements for a Personnel-Carrying Device (PCD) not specified herein, will be specified in the individual device specification.

1.2. Certification. LEAL certification approval is granted by the U.S. Army DEVCOM SC with concurrence from DEVCOM AvMC. Certification is confirmation of a successful assessment of the device against the requirements for being flown as a LEAL under a DoD aircraft. PCD to be certified for LEAL must meet the requirements of this standard. Requirements are established to ensure the live cargo can be repeatedly transported safely and will arrive at its destination safely. Evaluation for safe transport of live cargo may require analysis or testing above and beyond the guidance provided herein. DEVCOM AvMC SRD will identify the scope of evaluation required.

a. The requirements to meet other modes of transportation may be more restrictive than provided herein and will be met when applicable.

b. PCD with cargo storage areas must also meet the cargo tie-down criteria of MIL-STD-209.

c. PCD submitted for certification must be in their final fielding/fielded configuration. Physical design changes made to devices during, or post certification may invalidate testing results and require retesting.

1.3. Operation. Design criteria held within this standard are intended to ensure live cargo arrives at its destination safely. Suspension structure is a significant portion; however, all parts of the device will be evaluated against LEAL testing requirements and applies to the integrity of the entire system.

1.4. Applicability. This standard applies to new or modified devices.

a. All new developmental, non-developmental, and military-adapted commercial devices with LEAL requirements.

b. Previously qualified devices which have been modified when the modifications result in changes to device materials, retention features, weight, Center of Gravity location, lift provisions, supporting structure, or projected frontal area.

c. LEAL is applicable to all DoD aircraft.

1.5. Metric equivalents. Metric equivalents will conform to IEEE/ASTM SI 10.

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1.6. Exceptions. Integrated survival vest harnesses are considered Aviation Life Support Equipment and governed as such.

2. APPLICABLE PUBLICATIONS

2.1. General. The publications listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include publications cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2. Government publications.

2.2.1. Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

RR-C-271	Chains and Attachments, Carbon and Alloy Steel

(Copies of this publication is available online at https://assist.dla.mil)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-R-8592/3	Military Specification, Survivors, Sling, Rescue Helicopter Equipment.
MIL-PRF-8625	Anodic Coatings for Aluminum and Aluminum Alloy
MIL-DTL-23599	Detail Specification, Hook, Rescue
MIL-L-37957	Military Specification, Litter, Rigid, Stokes

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-209	Interface Standard for Lifting and Tiedown Provisions.
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-882	System Safety
MIL-STD-889	Galvanic Compatibility of Electrically Conductive Materials

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MIL-STD-913	Requirements for The Certification of Sling Loaded Military Equipment for External Transportation by Department of Defense Helicopters
MIL-STD-1472	Human Engineering

(Copies of these publications are available online at https://quicksearch.dla.mil)

2.2.2. Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

U.S. ARMY AEROMEDICAL RESEARCH LABORITORY

JECETS	Joint En Route Care Equipment Test Standard

(Copies of these documents are from United States Army Aeromedical Research Laboratory P.O. Box 620577 Fort Rucker, AL 36362.)

CODE OF FEDERAL REGULATIONS

Title 14 Part 29 Subpart D	Airworthiness Standards: Transport Category Rotorcraft, Design and Construction
Title 14 Part 29 Subpart C	Airworthiness Standards: Strength Requirements
Title 14 Part 27 Subpart D	Airworthiness Standards: Normal Category Rotorcraft, Design and Construction

(Copies of these publications are available online at https://www.ecfr.gov/ or from the Superintendent of Documents, U.S. Government Printing Office, North Capitol & "H" Streets, N.W., Washington, DC 20402-0002.)

2.3. Non-Government publications. The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

NORTH ATLANTIC TREATY ORGANIZATION

STANAG 3295	Horse Collar/Rescue Strop Type Hoisting Gear
STANAG 3542	Technical Criteria for the Transport of Cargo by Helicopter.

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(Copies of these publications are available online at https://nso.nato.int/nso/nsdd/main/list-promulg)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D6193	Standard Practice for Seams and Stitches
F2821	Standard Test Method for Basket Type Rescue Litters
F2491	Standard Guide for Determining Factors for Technical Rescue Equipment
F2266	Standard Specification for Masses Used in Testing Rescue Systems and Components

(Copies of these publications are available online at https://www.astm.org/products-services/standards-and-publications/standards.html or from ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA, 19428)

AMERCIAN WELDING SOCIETY

AWS D17.1/D17.M	Specification for Fusion Welding for Aerospace Applications
WHB	Welding Handbook

(Copies of these publications are available online at https://pubs.aws.org/ or from American Welding Society, 8669 NW 36 Street, #130 Miami, Florida 33166-6672)

SAE INTERNATIONAL

AMS2175	Castings, Classification, and inspection of
AMS2460	Plating, Chromium
AMS-QQ-P416	Plating, Cadmium (Electrodeposited)

(Copies of these publications may be purchased online at www.sae.org or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001)

PARACHUTE INDUSTRY ASSOCIATION

PS70087	Clevises, Suspension, Air Delivery-Type 1
PIA-DTL-40137	Clevises, Rigging, Air Delivery Type: General, Specification For

(Copies of these publications may be purchased online at www.pia.com or from Parachute Industry Association, 6499 S Kings Ranch Rd, Gold Canyon, AZ 85118)

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2.4. Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1. Ancillary Equipment. Equipment used during LEAL operations to alter loading or flight characteristics of the device. Gyroscopic stabilizers are an example of ancillary equipment.

3.2. Anthropometric data. Current anthropometric data will be referred to for occupant weights. Occupant weight for the Live Load calculation is derived from the ANSURII Database.

3.3. Attaching Hardware. Hardware used to connect a PCD to and suspend a device from the hoist hook, cargo hook, or airframe of an aircraft.

3.4. Center of Gravity (CG). The point at which the entire weight of a body may be considered as concentrated so if supported at this point the body would remain in equilibrium in any position.

3.5. Certification. The confirmation of requirements, design characteristics and flight performance to provide confidence in the PCD 's safe transport of live external loads.

3.6. Design Limit Load (DLL). The maximum applied dynamic force the device is design to experience during transport. DLL for PCD must meet or exceed a safety factor equal to the lift point load factor (see para 3.13) IAW Title 14 Part 29 subpart C. The DLL is used in proof-load testing. This is equivalent to the proof load or dynamic load.

3.7. Flight Evaluation. An evaluation during which a PCD(s) is rigged in its load configuration and flown through specific maneuvers by military aircraft. The evaluation is used to verify stability of the host aircraft and PCD during flight as well as to verify that the PCD(s) can withstand the dynamic forces induced by flight.

3.8. Lifting Line. A fixed length line used to attach a PCD to the aircraft attaching point.

3.9. Lift Point. The load bearing connection that transfers LEALWT from the PCD to aircraft.

3.10. Lift Point Load Factor (LF). A multiplier of the static load which results in the design limit load. The LF is equal to 3.5 multiplied by the maximum external load for which authorization IAW 14 CFR 27 Subpart D.

3.11. Lifting Provision. An integral part of the PCD, commonly called a lug, pad

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eye, ring, eye, or attachment. A lifting provision provides a means of attaching the PCD to the aircraft.

3.12. Live External Attached Load (LEAL). A specific method of insertion/extraction/transport of live occupants via a PCD that is attached externally to the aircraft. The potential attachment points for LEAL are the rescue hoist, cargo hook and airframe.

3.13. Live External Attached Load Weight (LEALWT). The specified weight (mass times gravitational acceleration) of the PCD and lifting line when prepared for transport with occupants or payload. Typically, LEALWT does not include aircrew weight and requires fuel level to be adjusted based on LEAL mission type. LEALWT is used to calculate the lift point load point factor.

3.14. Live Load. The total anticipated weight of all occupants being transported.

3.15. Load. A force applied to a structural member or attaching hardware.

3.16. Non-Standard Lifting Component. Any load bearing equipment that is required for rigging and lifting which does not have a DoD type classification such as per Army Regulation 700-142.

3.17. Occupant. Human cargo.

3.18. Occupant Weight. The weight of the occupant used in the Live Load calculation. The current weight of a 95th percentile civilian male is 250lbs IAW ANSURII Database.

3.19. Payload. Any material that is transported by the PCD, other than live loads which is not physically integrated. This includes, but is not limited to, cargo, mission equipment, and personnel gear.

3.20. Payload Tie Down Provisions. Locations connected to the primary structure of the PCD providing a location to secure payload.

3.21. Permanent Deformation. Any detectable permanent change in the original dimensions or shape of the provision or connecting structure remaining, following the removal of an applied force. Detection methods are subject to the location being checked for deformation.

3.22. Personnel-Carrying Device (PCD). Device that is either attached to a hoist, cargo hook, or mounted to the aircraft airframe during LEAL or hoist operations. These device(s) have the structural capability and features needed to transport occupants externally to the aircraft (e.g. a life safety harness with or without a quick release and strop with a connector ring, a rigid basket or a cage).

3.23. Pitch. The rotation of a body about an axis passing through it laterally.

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3.24. Plane of the Provisions. Geometric plane connecting the centers of groups of at least three lifting provisions on a PCD.

3.25. Powered Ascender. Remote powered device used to insert/extract personnel or PCDs.

3.26. Proof Load. See "Design Limit Load".

3.27. Proof Load Test. A test in which the lifting line and PCD structure are subjected to the DLL.

3.28. Restraint Design Load (RDL). The weight of a current 95th percentile civilian male multiplied by the LF (see paragraph 3.13).

3.29. Restraint Attachments. The hardware attachment of the occupant restraint to the PCD.

3.30. Secondary Safety Device (SSD). An independent means of attaching a PCD to an aircraft cargo hook. Used in conjunction with the primary attachment of a cargo hook attachable PCD and must meet the Upper Load Limit (ULL) requirements of the PCD.

3.31. Short Haul. The capability of transporting one or more occupants via a PCD the practical shortest distance.

3.32. Splice. A temporary hoist hook should the primary hook become damaged, fouled, or cut.

3.33. Static Lift Orientation (SLO). The orientation/configuration of the PCD as determined by the static lift test.

3.34. Static Lift Test. A test in which a PCD is suspended in the proposed, rigged configuration without movement. The purpose is to identify interferences, contact points, forward pitch, and other critical characteristics.

3.35. Static Load. The resultant force (measured or anticipated) imposed on a lifting provision and supporting structure under static lift test conditions with a PCD at its WLL.

3.36. Supporting Structure. Material or components that comprise the load path from the device's center of gravity to the lifting provisions (including the lifting provisions).

3.37. Ultimate Load Limit (ULL). The applied force (not less than the DLL multiplied by 1.5) the supporting structure can sustain without breaking, rupturing, or otherwise becoming unstable.

3.38. Working Load Limit (WLL). The maximum rated load capacity a PCD is

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authorized to handle.

3.39. Yield load. The force at which the supporting structure exhibits permanent deformation after the force is removed. The yield load must be greater than the DLL.

3.40. Acronyms.

ASSIST	Acquisition Streamlining and Standardization Information System
ASTM	American Society for Testing and Materials
AvMC	Aviation and Missile Command
CG	Center of Gravity
DEVCOM	Combat Capabilities Development Command
DLL	Design Limit Load
DoD	Department of Defense
HEC	Human External Cargo
LEALWT	Live External Attached Load Weight
LF	Lift Point Load Factor
PCD	Personnel-Carrying Device
RDL	Restraint Design Load
SC	Soldier Center
SLO	Static Lift Orientation
SRD	Systems Readiness Directorate
SSD	Secondary Safety Device
ULL	Ultimate Load Limit
WLL	Working Load Limit

4. GENERAL REQUIREMENTS

4.1. Lifting Provisions. All PCD s shall have provisions or structure identified for lifting IAW MIL-STD-209. Lifting provisions shall be affixed in a permanent manner. Tethering cables do not meet this requirement.

4.2. Payload Tiedown Provisions. Whenever possible, if so equipped, payload tiedown provisions should be permanently mounted to the PCD. Any removable parts should have a designated stowage area on the PCD and marked as being part of the PCD.

4.3. Hardware Properties. All lifting and payload tied down provision hardware shall meet the requirements of PS70087 and RR-C-271.

4.4. Freezing and Jamming. All lifting and payload tiedown provisions shall meet the freezing and jamming requirements of MIL-STD-209.

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4.5. Operating condition. PCD shall be able to withstand the following environments IAW MIL-STD-810.

4.5.1. Temperature. PCD shall operate at any ambient temperature between –65 °F to +160 °F.

4.5.2. Sand. PCD shall withstand blowing sand environment.

4.5.3. Salt Spray. PCD shall withstand salt spray environment.

4.5.4. Shock. PCD shall withstand functional shock environment.

4.5.5. Water immersion. PCD shall withstand saltwater immersion without permanent damage or functional impairment.

4.5.6. Fungus. PCD shall withstand the fungus environment.

4.6. Sharp Corners. Lifting and payload tiedown provisions shall not have sharp corners or edges. All edges shall have a minimum 1/8" radius round or 1/16" x 1/16" chamfer.

4.7. Direction of Flight. Direction of flight will be determined by static lift, proof load testing and flight evaluation. When system specific components may be at risk of damage from pickup, set-down, or dynamic pressure, the vendor shall identify a critical direction of flight for testing and evaluation.

4.8. Airspeed and Dynamic Pressure. All aspects of the PCD should be designed to withstand dynamic pressures experienced during flight. Components will experience dynamic pressures on exposed surface areas from the relative wind. Analysis and testing shall be done to validate PCD design. During suspension, a PCD may rotate during flight. The analysis should include the worst-case wind angle of incidence. Design integrity shall be demonstrated during a PCD flight evaluation.

4.9. Personnel Carrying Device (PCD). All PCD s shall be specifically approved by DoD components for transport of LEAL.

4.9.1. PCD s will be assigned to one of three categories.

a. Hoist Attachable. Lifted via hoist hook and are designed to transport occupants in the horizonal, vertical or seated position. In addition to the criteria outlined in this standard hoist attachable PCD s -whether in their entirety or as part of a modular system- may also be required to conform to MIL-L-37957, MIL-R-8592/3, STANAG 3295, PIA-DTL-40137, or ASTM F2821.

b. Cargo Hook Attachable. Lifted via the cargo hook and are designed to transport occupants in a simultaneous combination of supine, standing, or seated position.

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c. Airframe Attachable. Typically lifted via a direct connection to the airframe and are designed to transport occupants in a simultaneous combination of supine, standing, or seated position.

4.9.2. In addition to the criteria outlined in this standard, all PCD manufactures shall adhere to ASTM F2491.

4.10. SSD. All PCD attached to an aircraft cargo hook that is not rated for Human External Cargo (HEC) must be secured with an SSD. In addition, the SSD must have an independent means of quick release. The SSD activation means shall be accessible for manipulation within 30 seconds if required.

4.11. Non-Lifting structure. All components of the PCD shall be evaluated to determine their required adherence to this standard.

4.12. Textiles. Load-bearing textile materials used in the construction of PCD s shall be made from virgin, synthetic, continuous filament fiber. All webbing ends shall be secured by heat sealing or by another method that prevents unraveling. Stitching shall meet or exceed ASTM D6193. Exterior components of the PCD constructed from textiles shall be attached to the PCD in such a manner as to prevent inadvertent separation. Permanently mounted access panel covers or pouches shall have a means of being properly secured.

4.13. Metals. Load-bearing hardware components of PCD s shall be constructed of forged, machined, stamped, extruded, or cast metal.

4.13.1. Castings shall meet Class I, Grade A requirements of SAE-AMS2175, Castings, Classification, and Inspection of, and Title 14 Part 29 subpart D, Casting Factors.

4.13.2. Welding shall be durable, sound, and in accordance with the applicable recommendation of the Welder's Handbook, American Welding Society and AWS D17.1.

4.13.3. Acceptable welds shall have no cracks or voids, and thorough fusion shall exist between adjacent layers of weld metal and base metal.

4.13.4. After welding, any projections, loose weld spatter or other defects which could possibly result in injury shall be removed.

4.13.5. A fatigue tolerance evaluation shall be completed IAW 14 CFR 29.571.

4.13.6. All metals shall be of the corrosion resistant type or shall be treated to resist corrosion in accordance with MIL-PRF-8625F, AMS2460B or AMSQQP416G. Dissimilar metals shall not be used in accordance with MIL-STD-889.

4.14. Finish. The finishing of components, as specified, shall be smooth, evenly applied and adherent. There shall be no voids, discontinuities in finish or damages finish areas.

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4.15. Rigging Procedures.

4.15.1. Lifting Hardware. Lifting hardware as defined above is the only hardware authorized to carry LEAL on a DoD aircraft. Miscellaneous hardware required for specific loads must be approved by DEVCOM AvMC and receive concurrence from DEVCOM SC during the formal testing process. Powered ascenders and splices shall meet the requirements of lifting hardware.

4.15.2. Line Contact - LEAL. There is no minimum clearance required for LEAL. No part of the PCD shall be allowed to contact a textile portion of the lifting line. Any points of contact between the PCD and lifting line shall be designed to withstand the compressive loading applied to it.

4.15.3. Minimum Load Lift Angle. No section of a PCD shall have a load lift angle of less than 40°.

4.15.4. Non-Standard Lifting Components. Non-standard lifting components are not permitted unless approved by DEVCOM-AvMC. Operation and use must be specified in the equipment manuals. The non-standard lifting components shall have a storage location in such that it will always be present and available for use. Examples of non-standard lifting components are a quick splice or a powered ascender.

4.15.5. Lifting Line Length. Based on the PCD, mission, and the intended aircraft, line length will be adjusted to meet the requirements of this standard.

4.15.6. Climbing on PCD s. PCD height and configuration may require personnel to climb upon it to access lift provisions and to connect lifting line to the aircraft. In such cases, there shall be a path/method identified to permit personnel to safely ascend/descend the PCD.

4.16. Testing Configuration. All testing shall be performed to simulate how the device is meant to be transported and operationally used. The mass shall be positioned to match the actual payload. Masses used for testing shall be IAW ASTM F2266. DEVCOM AvMC will approve all simulated configurations and deviations prior to testing.

4.17. Environmental Testing. Environmental testing of the PCD shall be conducted IAW MIL-STD-810.

4.18. Identification. All PCD shall be clearly labeled with the following information

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in a manner that prevents removal, deformation, or fading.

a. Manufacturer's name
b. Country of manufacture
c. Contact number
d. Manufacturer's product identification
e. Serial number
f. Date of manufacture
g. Proof load rating
h. Working Load Limit

4.19. Safety Restraints. All PCD s shall have a means to secure occupants as to prevent inadvertent departure from the PCD. Method of restraint shall be in the form of an approved personal tether, harness, belt, or physical barrier such as a wall or railing (see 5.8. of this standard).

4.20. Continued Airworthiness. To ensure continued airworthiness, maintenance manuals shall specify the frequency, extent, and methods of inspection, along with corrosion prevention techniques for critical structures, systems, and components. They must also detail specific, substantiated repair limits and required maintenance procedures.

4.21. Human Systems Integration. PCD shall adhere to the human factors design criteria referenced in MIL-STD-1472

5. DETAILED REQUIREMENTS

5.1. Weight Restrictions. LEAL subjected devices must be within the capabilities of the desired lifting aircraft and attaching point.

5.2. Lift Provisions.

5.2.1. Number. The number of lift provisions shall be determined by satisfying the following strength requirements for all individual lifting lines and lift provisions:

a. Static Load Requirement: The calculated tension on any single component under static lift conditions, with the PCD loaded to its maximum Working Load Limit (WLL), shall not exceed the WLL of that component.

b. Dynamic Load Requirement: The entire lifting system shall be designed so that when subjected to the Design Limit Load (DLL), no part of the lifting assembly is stressed beyond its own DLL.

5.2.2. Location. The lift provisions shall be centralized or spaced as far apart as practical. The PCD CG shall be within the area bounded by the lift provisions as viewed from above (plan view). Lift provisions shall be positioned such that the maximum vertical reaction force is no greater than 1.2 times the minimum vertical reaction force across all lifting

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provisions. Refer to STANAG 3542.

5.2.3. Dimensions. Whenever practicable, lift provisions shall meet the dimensional interface requirements of MIL-STD-209 or MIL-DTL-23599. At minimum, the provisions must be sized to allow the lowest rated attaching hardware to smoothly interface. Cargo and hoist hooks shall pass through freely. "Freely" shall be with no need to feed, rattle, or manually guide.

5.2.4. Strength. The DLL of each lift provision shall be established by its share of the load as measured within the SLO.

5.2.5. Frame Attachment. Frame or other structural members may be used as lift points when they meet all the related criteria for lift provisions and are specifically designated and marked accordingly. The phrase "lift provision" shall include designated frame attachment points.

5.2.6. Validation. Lift provisions shall be validated during static lift and proof load testing and ULL validation.

5.2.7. Marking. Lift provisions for external aircraft transport shall adhere to marking requirements detailed in MIL-STD-209.

5.2.8. Ancillary Hardware. Hardware not permanently affixed to the PCD shall be defined in the new equipment specification and shall have a designated stowage location on the PCD. The specification shall provide guidance on the proper operation, installation, and stowage of the hardware. Such hardware must always remain with the PCD.

5.3. Static Lift Testing. Static lift testing consists of lifting the PCD in the proposed rigging configurations to determine lifting line lengths, line tensions, and contact point reaction loads. Static lift testing also determines how the PCD and lifting line will interact. Additionally, rigging personnel mounting/dismounting concerns, snag possibilities, and potential impact areas are identified. Static lift testing is typically the first exposure the certifying engineers have to the PCD. Preliminary rigging procedures are established during static lift testing.

5.3.1. Frequency. Static lift testing is performed on every new PCD submitted for certification.

5.3.2. Static stability. PCD s shall maintain stability while suspended in the rigged configuration as deemed acceptable by the certifying agency.

5.3.3. Aircraft Restrictions. All PCD configurations shall meet any limits and restrictions imposed by the intended lifting aircraft.

5.3.4. Instrumentation. At a minimum, load cells shall record forces at the apex of

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the PCD. This data is required for establishing the proof load criteria.

5.3.5. Results. Apex loads, lifting line lengths, line tensions, and flight attitude shall be recorded. Lifting line length shall be recorded by physical length. Additional data shall be noted as required.

5.3.6. Exception. DEVCOM AvMC with concurrence from DEVCOM SC, will determine if static lift testing can be waived.

5.4. Proof Load Testing. Proof load testing consists of a static pull of the PCD in the rigging configuration at the DLL to validate the load path from the CG to the lift provisions. Test results shall verify that lift provisions, contact points, and other structural components can support the DLL. Additional proof load tests may be required on the PCD 's individual components.

5.4.1. Loads. All loads measured and identified during static lift test which result from lifting line contact shall be multiplied by the LF. Resultant line tensions shall not exceed the DLL for the proposed lifting line.

5.4.2. Direction. The application of all proof-load forces shall be in the directions determined during from SLO.

5.4.3. Structural load path. Proof-load testing is to validate the PCD 's supporting structure. The test setup shall, as closely as possible, reflect the path the loads will propagate from the PCD 's CG to the lift provisions.

5.4.4. Test Apparatus. Proof-load testing is most simply performed by replacing the attaching hardware used for static lift with chains or wire rope and duplicating the lift configuration. The chains or wire rope shall be selected based on the maximum expected line tension during proof-loading, shall be within +/- 3 inches of the line length, and shall fit through all guides and lift provisions. Additionally, test apparatus may be arranged to match SLO angles within +/- 2°.

a. When appropriate, hooks will match the attaching hardware in dimension and meet the increased loading strength required by the test.

b. Achieving the DLL may be accomplished by adding weights to the PCD in appropriate areas, by lashing the PCD 's frame structure to the floor, or a combination of both. For PCD s such as shelters and containers, the most correct loading may involve adding mass to the internal floor. Localized damage from the mass may occur to the flooring without resulting in a proof-load test failure.

c. All lift provisions should be tested simultaneously. In circumstances when this is not practical, the test set up must reflect the appropriate support provided by the unused lift provisions.

5.4.5. External payload stowage areas. Payload stowage areas shall provide

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downward restraint of no less than the rated load limit of the stowage area multiplied by the LF.

5.4.6. Test Angle. The test setup shall orient the PCD within 1° of the recorded SLO pitch angle.

5.4.7. Contact Points. All points identified during the static lift where the lifting line encounters the PCD must be duplicated. The test angle shall be adjusted within its tolerance limit to ensure the contacts are achieved. No hardware or instrumentation used during the proof-loading shall create additional contact points or be chosen to better distribute the load.

5.4.8. Bearing Stresses. Artificial bearing stresses shall be avoided.

a. At lift provisions and contact points, the interfacing proof-load testing hardware shall be of similar size to the intended lifting line interface hardware.

b. Tiedowns for creating the proof-load shall be of sufficient width at contact points to minimum localized damage. In some instances, dunnage may be allowed to distribute the load.

5.4.9. Duration. All proof-load tests shall be applied for a minimum of 90 seconds.

5.4.10. Application. Forces applied during proof-load testing shall not be less than or exceed 10% greater than the calculated DLL throughout the test duration. (+10%/-0%)

5.4.11. Criteria. Proof-loads are successful when there is no permanent deformation to any of the structure being subjected to loading. Permanent deformation shall be determined by instrumented and visual inspections.

5.4.12. Instrumentation. At a minimum, load cells shall record forces at the apex of the lifting provisions. Load cell length shall be considered when simulating the static lifting line lengths. Additional data shall be captured as required.

5.4.13. Results. Proof-loading shall be assessed as a pass/fail evaluation. Measurements acquired to validate the criteria shall be permanently recorded. If permanent deformation is identified, follow-on inspections and tests shall be conducted to determine the cause of the deformation such as slippage of bolted joints, yielding of parts, or other. AvMC may accept a “fail” condition as acceptable based on the identified cause of the deformation.

5.4.14. Exception. DEVCOM-SC with concurrence from DEVCOM AvMC will determine if proof-load testing can be waived.

5.5. Flight evaluation. A flight evaluation is performed to identify any rigging or stability issues which may be encountered in the field. Prior to the flight, the rigging procedure initiated during static lift will be finalized.

5.5.1. Pre-flight rigging. Preparation for the flight evaluation shall not be a timed

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event as the rigging procedures will not be fully established.

5.5.2. Configuration. The certifying agency shall identify the critical configuration(s) for the flight evaluation.

5.5.3. Flight pattern. The certifying agency shall identify which maneuvers are required to be performed for evaluation.

5.5.4. Aircraft restrictions. All maneuvers flown shall be performed without exceeding flight manual restrictions.

5.5.5. Dynamic stability. The stability of the PCD and the stability of the host aircraft should be evaluated for each maneuver performed.

5.5.6. Criteria. The PCD and host aircraft shall not sustain any damage and shall be mission capable upon completion of the flight.

5.5.7. Results. Evaluative ratings shall be assigned to the behavior of the load during each of the flight maneuvers. The flight evaluation shall establish flight limits for airspeed and angle of bank at which the PCD can be safely carried.

5.5.8. Exception. DEVCOM SC with concurrence from DEVCOM AvMC will determine if flight evaluation can be waived.

5.6. Modified Personnel Carrying Devices (PCD). A PCD which has been modified from the condition it was previously certified in shall be evaluated for resubmission to the tests noted in this standard.

5.7. Ultimate load limit (ULL). The ultimate load limit of each tiedown provision and supporting structure shall be at least 1.5 times the design load limit. ULL of all structural components in the load path shall be proven by analysis or testing.

5.8. Occupant Safety

5.8.1. Restraints.

a. Restraints shall be designed in accordance with MIL-R-8592/3 and tested IAW JECETS.

b. Restraints shall be designed to prevent the occupant from inadvertent departure from the PCD.

c. Number of restraints: There shall be one restraint per occupant.

d. Sizing and interchangeability: shall be adjustable or interchangeable for larger and smaller civilian populations (i.e. adult male/female and children) in accordance with the current anthropometric data (e.g., chest / abdomen circumference, shoulder width, etc.).

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e. Restraint Attachments: Restraints shall have hardware to connect to the PCD.

f. Restraints shall be easy to don/doff and secure.

g. Restraints shall have a method for safe stowage as to prevent injury or ingress/egress hazard when not in use.

h. Restraint and Restraint Attachment Performance. The minimum strength of the restraint components shall be equal to or greater than the RDL.

5.8.2. Occupant Orientation. The PCD must demonstrate restraint safety for all intended occupant orientations including supine, seated, and standing positions during transport, as approved by the certifying agency.

5.8.3. Ingress/Egress.

a. The PCD must be stable for safe ingress/egress and during transport as deemed acceptable by the certifying agency.

b. Occupant must be capable of ingress/egress from the aircraft if the PCD is a singular wearable device such as a harness.

c. PCDs must have procedures for ingress/egress depending on occupant position (i.e., supine, seated, standing, or other such as a child sitting on a parent's lap).

5.8.4. Identification. Restraints shall be labeled IAW section 4.18. of this standard and any applicable restraint-specific identification requirements.

5.9. Safety. A system safety assessment shall be conducted in accordance with MIL-STD-882E or an accepted equivalent process to identify, assess, mitigate, and document LEAL hazards, including inadvertent release, occupant departure, mis-rigging, emergency jettison, aircraft interface hazards, human factors hazards, and maintenance-related hazards. Prior accepted system safety assessments may be credited when applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1. Intended use. This standard covers design, testing and performance requirements of PCD s for external transport by all DoD aircraft.

6.2. Certification by Similarity. This is a process by which a PCD is certified without the full scope of testing. Certification by Similarity is a path chosen by the certification engineering team during analysis. A PCD is compared to previously certified

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PCD s and evaluated to determine if the physical characteristics are similar enough to waive testing. The result of the comparison may be the elimination of a single test or the entire set of tests. Certification by Similarity should not be directly requested. It is a cost savings measure and will be considered for every program. Additionally, thew U.S. Army may recognize other airworthiness authorities (such as civilian or Military Airworthiness Authorities of another Nation).

6.3. Issue of ASSIST. When this standard is used in acquisition, the applicable issue of ASSIST must be referenced in the solicitation (see 2.2.1.).

6.4. Waivers. Waivers to this standard regarding LEAL may be granted by DEVCOM AvMC with concurrence from DEVCOM SC.

6.5. International standardization agreements. Certain provisions of this standard are the subject of international standardization agreements. When change notice, revision, or cancellation of this standard is proposed that will modify the international agreements concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.6. Subject term (key word) listing.

Aircraft
Airworthiness
Certification
Design Limit Load (DLL)
Flight Evaluation
Hardware
Lifting Provisions
Live External Attached Load (LEAL)
Payload
Personnel-Carrying Device (PCD)
Proof Load Test
Restraints
Rigging
Static Lift Test
Ultimate Load Limit (ULL)
Working Load Limit (WLL)

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APPENDIX A

Adopting Government Agencies

SCOPE This Appendix is not a mandatory part of the standard. The information contained herein is intended for DoD offices adopting the use of MIL-STD-3088.

US Army:

US Marine Corp:

US NAVY:

US Coast Guard:

US Army National Guard:

US Air National Guard:

US Air Force:

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CONCLUDING MATERIAL

Custodians:
Army – AV

Preparing activity:
Army - AV
(Project 4240-2026-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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